

BLTK1 Murine Leydig Cells: A Novel Steroidogenic Model for Evaluating the Effects of Reproductive and Developmental Toxicants

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Leydig cells are the primary site of androgen biosynthesis in males. Several environmental toxicants target steroidogenesis resulting in both developmental and reproductive effects including testicular dysgenesis syndrome. The aim of this study was to evaluate the effect of several structurally diverse endocrine disrupting compounds (EDCs) on steroidogenesis in a novel BLTK1 murine Leydig cell model. We demonstrate that BLTK1 cells possess a fully functional steroidogenic pathway that produces low basal levels of testosterone (T) and express all the necessary steroidogenic enzymes including *Star*, *Cyp11a1*, *Cyp17a1*, *Hsd3b1*, *Hsd17b3*, and *Srd5a1*. Recombinant human chorionic gonadotropin (rhCG) and forskolin (FSK) elicited concentration- and time-dependent induction of 3',5'-cyclic adenosine monophosphate, progesterone (P), and T, as well as the differential expression of *Star*, *Hsd3b6*, *Hsd17b3*, and *Srd5a1* messenger RNA levels. The evaluation of several structurally diverse male reproductive toxicants including 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD), atrazine, prochloraz, triclosan, monoethylhexyl phthalate (MEHP), glyphosate, and RDX in BLTK1 cells suggests different modes of action perturb steroidogenesis. For example, prochloraz and triclosan antifungals reduced rhCG induction of T, consistent with published *in vivo* data but did not alter basal T levels. In contrast, atrazine and MEHP elicited modest induction of basal T but antagonized rhCG-mediated induction of T levels, whereas TCDD, glyphosate, and RDX had no effect on basal or rhCG induction of T in BLTK1 cells. These results suggest that BLTK1 cells maintain rhCG-inducible steroidogenesis and are a viable *in vitro* Leydig cell model to evaluate the effects of EDCs on steroidogenesis. This model can also be used to elucidate the different mechanisms underlying toxicant-mediated disruption of steroidogenesis.

Key Words: steroidogenesis; Leydig cells; endocrine disruptors; endocrine toxicology.

Steroidogenesis is a potential target for drugs, chemicals, natural products, and environmental contaminants that adversely impact reproductive development and fertility (Sanderson, 2006;

Whitehead and Rice, 2006). The mechanisms involved in altering steroid levels remain largely unknown but could involve disrupting enzyme expression and/or activity that affects testosterone (T) biosynthesis (Vinggaard *et al.*, 2006). For example, fungicides, pesticides, and phthalates elicit reproductive tract abnormalities, compromise reproductive fitness, and cause cancer as a result of altering steroid levels (Gray *et al.*, 2006; Scott *et al.*, 2009; Vinggaard *et al.*, 2006; Wilson *et al.*, 2008). Developmental exposure to high doses of endocrine disrupting compounds (EDCs) has been associated with testicular dysgenesis syndrome in humans, comprised of adverse effects including cryptorchidism, hypospadias, low sperm count, and testicular cancer, such effects have been characterized in rodents and wildlife species with similar etiology (Edwards *et al.*, 2006; Sharpe, 2006; Skakkebaek *et al.*, 2001; Wohlfahrt-Veje *et al.*, 2009).

Current approaches to identify developmental and reproductive toxicants involve examining offspring after dosing pregnant animals to maternally toxic levels (Kimmel *et al.*, 1993). Apical endpoints such as nipple retention, anogenital distance, vaginal opening, and balano-preputial separation are assessed in addition to evaluating reproductive fitness (Cooper, 2009; Marty *et al.*, 2009). To comply with the European Commission's Registration, Evaluation, Authorization, and Restriction of Chemicals initiative, approximately \$2.9 billion and 41 million animals are required for reproductive toxicity testing alone using current approaches (Breithaupt, 2006; Brown, 2003; Hartung and Rovida, 2009; Hofer *et al.*, 2004). Such animal-based testing is expensive, time consuming, generally descriptive, poses ethical concerns and does not significantly contribute to the elucidation of mechanisms (Andersen and Krewski, 2009; National Research Council, 2007; Daston, 2007; Daston *et al.*, 2010; Holsapple *et al.*, 2009; Knudsen and Daston, 2010). Furthermore, they are contrary to the replacement, reduction, and refinement of animal testing. Alternative short-term assays, such as the sliced testes assay and the human H295R adrenal

corticocarcinoma cell-based assay used in tier 1 of the Endocrine Disruptor Screening Program, still require animals or are based on adrenal steroidogenesis, which is not the primary site of sex hormone production (Borgert *et al.*, 2011; Harding *et al.*, 2006; Hecker *et al.*, 2006; Higley *et al.*, 2010). Therefore, *in vitro* gonadal-based steroidogenesis assays that support ranking and prioritization for further investigation are needed.

The "Toxicity Testing in the 21st Century: A Vision and a Strategy" report proposed testing should be less reliant on whole animal studies with more effort devoted to systems oriented computational models developed using *in vitro* and *in vivo* data that could screen chemicals, metabolites, and mixtures for potential toxicity (National Research Council, 2007). In response, the U.S. Environmental Protection Agency (USEPA), the National Institute of Environmental Health Sciences National Toxicology Program, and the U.S. Food and Drug Administration have initiated ToxCast, Tox21, and EDP21 programs in collaboration with the National Institute of Health Chemical Genomics Center, as well as other high-throughput service providers, to use *in silico* and *in vitro* assays to screen chemicals for potential toxicity (Dix *et al.*, 2007; Shukla *et al.*, 2010; USEPA, 2011). These programs use informatic platforms to integrate data from thousands of chemicals that are robotically screened in diverse assays to develop predictive computational models of toxicity. Similarly, the European Union has established ChemScreen and ReProTect to develop simple rapid *in vitro/in silico* screening systems for reproductive toxicants (www.chemscreen.eu; www.reprotect.eu; Schenk *et al.*, 2010). Pathways and overrepresented gene functions identified in dose-response genomic and high-throughput screening studies may also be used to determine points of departure in provisional peer reviewed toxicity values (<http://hhpprtv.onml.gov>) for chemicals not comprehensively evaluated by the Integrated Risk Information System (<http://www.epa.gov/iris/index.html>) or where there is minimal safety data. Despite this broad spectrum of programs, which includes assays for cell viability, apoptosis, mitochondrial function, receptor-mediated activity, and DNA damage in human, mouse, and rat cell lines, as well as cell-free systems, none assesses steroidogenesis in a high-throughput format (USEPA, 2007).

In mammals, the induction of steroidogenesis has been thoroughly studied. Luteinizing hormone (LH) secreted from the anterior pituitary induces gonadal steroidogenesis. LH binds G-protein-coupled LH-human chorionic gonadotropin (hCG) receptors (LHCGR) on the surface of Leydig cells. Subsequent dissociation of the G_{α_s} subunit stimulates adenylyl cyclase activity producing 3',5'-cyclic adenosine monophosphate (cAMP) that induces gene expression and enzyme activity increasing steroid biosynthesis. BLTK1 cells, isolated from a testicular tumor that developed in a transgenic mouse expressing the mouse inhibin α promoter/simian virus 40 T-antigen fusion gene (Kananen *et al.*, 1996; Rahman and Huhtaniemi, 2004), retain functional LHCGR-mediated steroidogenesis. They are stable over multiple passages, easily transfected, and exhibit

excellent growth characteristics. In this study, recombinant human chorionic gonadotropin (rhCG) and forskolin (FSK) are used as positive controls for the induction of steroidogenesis, as measured by increases in progesterone (P), T, and 17 β -estradiol (E2) levels in media. We investigate murine BLTK1 Leydig cells as a novel model for evaluating the effects of chemicals on steroidogenesis. Our results demonstrate that BLTK1 cells can be used to screen substances that alter intracellular cAMP, steroidogenic gene expression, and sex steroid levels.

MATERIALS AND METHODS

Cell culture and treatment. Mouse Leydig BLTK1 (BLT-1 cells, clone K1) cells were isolated from a testicular tumor that developed in a transgenic mouse expressing the mouse inhibin α promoter/simian virus 40 T-antigen fusion gene (Rahman and Huhtaniemi, 2004). Cells were maintained in phenol red-free Dulbecco's modified Eagle medium nutrient mixture F-12 (DMEM/F-12 media) (Invitrogen, Carlsbad, CA) with 10% fetal bovine serum (Thermo Scientific HyClone, Logan, UT), 100 U/ml penicillin and 100 μ g/ml streptomycin (Invitrogen) and incubated under standard conditions (5% CO₂, 37°C). For the evaluation of steroidogenic enzyme and receptor expression, cells were grown to 80% confluency and harvested without any treatment. For 3',5'-cyclic adenosine monophosphate (cAMP), progesterone (P), testosterone (T), and estradiol (E2) determination, cells were grown to 80% confluency, transferred into 24-well tissue culture plates (Sarstedt, Newton, NC), and incubated overnight. Cells were treated with dimethyl sulfoxide (DMSO) (Sigma, St Louis, MO) vehicle, or 0.1, 0.3, 1, 3, 10, 30, or 100 ng/ml rhCG (obtained from A.F. Parlow, National Institute of Diabetes and Digestive and Kidney Disease's National Hormone & Peptide Program, Harbor-UCLA Medical Center, Torrance, CA) or 0.1, 0.3, 1, 3, 10, 30, or 100 μ M FSK (Sigma), and media were collected at indicated times. Time course studies were conducted with DMSO vehicle, 3 ng/ml rhCG, or 10 μ M FSK, and media were collected at 1, 2, 4, 8, 12, 24, or 48 h. Gene expression studies used the same study design, concentrations, and time points with cells seeded into T-25 flasks (Sarstedt).

MTT assay. BLTK1 cells in 96-well plates (Sarstedt) were treated with 1, 3, 10, 30, 100, 300, or 600 ng/ml rhCG (positive control), 1, 3, 10, 30, 100, 300, or 600 μ M FSK (positive control), or 1, 3, 10, 30, 100, 300, or 600 μ M of test compound in triplicate, respectively. Media were aspirated after 24 h and replaced with 50 μ l of fresh MTT reagent (5 mg/ml thiazolyl blue tetrazolium bromide [Sigma] in PBS). Following a 3 h incubation, MTT reagent was aspirated and replaced with 150 μ l DMSO. Cells were incubated for 2 h followed by absorbance measurements at 595 and 650 nm (A595–A650) using an Emax precision microplate reader (Molecular Devices, Sunnyvale, CA). Results are reported as percentage of control calculated from the relative absorbance of treated versus DMSO controls where 100% indicates no cytotoxicity.

Western blotting. Lysates were prepared in lysis buffer (50mM Tris-HCl, pH 7.4, 150mM NaCl, 1mM EDTA, 0.5% Triton, and protease inhibitor tablet [Roche Diagnostics, Indianapolis, IN]). Protein concentration was determined spectrophotometrically by Bradford assay (Bio-Rad Laboratories, Hercules, CA). Extracts (50 μ g) were resolved on a 10% SDS/polyacrylamide gel and transferred to Immobilon membrane (Millipore, Billerica, MA). Membranes were incubated with 1 μ g/ml primary antibody (rabbit anti-StAR [Santa Cruz Biotechnology, Santa Cruz, CA; Catalog No. 25806], rabbit anti-Cyp11a1 [Millipore; Catalog No. AB1244], goat anti-Cyp17a1 [Santa Cruz; Catalog No. SC-46081], goat anti-3 β HSD [Santa Cruz; Catalog No. SC-30820], goat anti-Srd5a1 [Santa Cruz; Catalog No. SC-20399], and rabbit anti-Cyp19a1 [Abcam, Cambridge, MA; Catalog No. ab51924]) at 4°C overnight. Membranes were washed and incubated with donkey anti-rabbit or donkey anti-goat IgG conjugated with horseradish peroxidase (Santa Cruz; Catalog No. SC-2077 and SC-2056, respectively) for 1 h. Enhanced Chemiluminescence (Thermo Scientific Pierce, Rockford, IL) was used to visualize detected protein.

Enzyme immunoassays. Enzyme immunoassay (EIA) kits for T (Cayman Chemical Company, Ann Arbor, MI; limit of detection 6 pg/ml, cross-reactivity 27% for 5 α -dihydrotestosterone and 19% for 5 β -dihydrotestosterone), E2 (Cayman Chemical Company; limit of detection 20 pg/ml, cross-reactivity 14% for estradiol-3-glucuronide, 12% for estrone, and 10% for estradiol 17-glucuronide), and P (ALPCO Diagnostics, Salem, NH) were used according to the manufacturer's instructions. Intracellular cAMP was measured by EIA (Cayman Chemical Company; limit of detection 3 pmol/ml cAMP, cross-reactivity 1.5% for cyclic guanosine monophosphate) following hydrochloric acid extraction. All kits consisted of 96-well precoated antibody plates, for which samples compete for binding with conjugated hormone. Following incubation, plates were washed and measured at 420 nm for cAMP, T, and E2 or 405 nm for P, using an Emax precision microplate reader (Molecular Devices). Standards (0.3–750 pmol/ml cAMP, 0–60 ng/ml P, 3.9–500 pg/ml T, or 6.6–4000 pg/ml E2) were used to generate a standard curve for quantification.

RNA isolation and gene expression. Total RNA was extracted from cell pellets using RNeasy Mini Kits (Qiagen, Valencia, CA) according to the manufacturer's protocol with an additional RNase-free DNase (Qiagen) digestion. RNA was quantified at 260 nm (A_{260}) and purity assessed using the A_{260}/A_{280} ratio, as well as by denaturing gel electrophoresis. First-strand complementary DNA (cDNA) was synthesized from RNA (1 μ g) using SuperScript II reverse transcriptase (Invitrogen) and anchored oligo-dT primer (Invitrogen) as described by the manufacturer. For real-time PCR (RT-PCR) evaluation of steroidogenic enzyme and receptor expression, the cDNA was used as template for PCR amplification with gene-specific primers (Supplementary table 1). Quantitative RT-PCR (QRT-PCR) was used to quantify concentration- and time-dependent expression of specific genes. Reactions in 96-well plates consisted of 30 μ l, including 1 μ l of cDNA template, 0.1 μ M forward and reverse gene-specific primers, 3mM MgCl₂, 1.0mM deoxynucleotide triphosphates, 0.025 IU AmpliTaq Gold, and 1 X SYBR Green PCR Buffer (Applied Biosystems, Foster City, CA) using an Applied Biosystems PRISM 7500 Sequence Detection System. Dissociation curve analysis assured single

product amplification. To control for differences in RNA loading, quality, and cDNA synthesis, samples were standardized to the geometric mean of three housekeeping genes: *ActB*, *Gapdh*, and *Hprt* (Vandesompele *et al.*, 2002). Results were quantified using a standard curve generated on the same 96-well plate and amplified by using purified cDNA product as template specific for each gene (serial 10 $\times$ dilutions from 10⁸ to 10¹ copies). The slope of the standard curve was used to assess amplification efficiency as described by the manufacturer with all amplification efficiencies > 90%. Fold changes were calculated relative to time-matched vehicle. Relative expression was scaled such that time-matched vehicle control expression equaled one for graphing purposes.

Dose-response modeling and statistical analyses. The ToxResponse modeler uses particle swarm optimization to identify the best fit across five model classes: sigmoidal, exponential, linear, quadratic, and Gaussian (Burgoon and Zacharewski, 2008). The best fitting model was then used to calculate half maximal effective concentration (EC₅₀) values. All statistical analyses were carried out using SAS v9.1 (SAS Institute, Cary, NC) by ANOVA, with Dunnett's or Tukey's *post hoc* tests for concentration-response and time course data, respectively. Differences between treatment groups were considered significant when $p < 0.05$ relative to time-matched DMSO control.

RESULTS

Steroidogenic Enzyme Expression in BLTK1 Cells

Steroidogenesis involves the conversion of cholesterol to progestogens (pregnenolone, progesterone [P], 17 α -hydroxypregnenolone, and 17 α -hydroxyprogesterone), followed by further metabolism to androgens (dehydroepiandrosterone [DHEA], androstenedione, testosterone [T], and 5 α -dihydrotestosterone) and estrogens including estrone and 17 β -estradiol (Fig. 1).

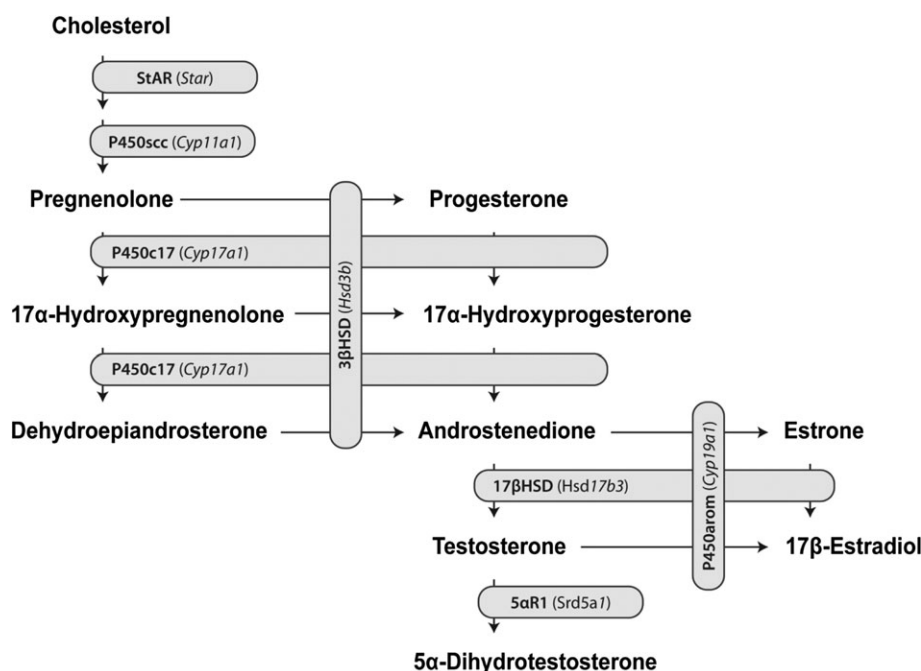


FIG. 1. The steroidogenic pathway. Steroid hormones are biosynthesized from cholesterol. Cholesterol is mobilized and delivered to the inner mitochondrial membrane by StAR, where it is cleaved by P450scc to produce pregnenolone. Subsequent enzymatic events in the smooth endoplasmic reticulum include metabolism by P450c17 and 3 β HSD to produce progestogens (i.e., progesterone (P), 17 α -hydroxypregnenolone, and 17 α -hydroxyprogesterone). Androgens (i.e., dehydroepiandrosterone and androstenedione) are produced by P450c17, then converted to testosterone (T) by 17 β HSD, and finally reduced to 5 α -dihydrotestosterone (DHT) by 5 α R1. Androgen metabolism by P450arom also yields estrogens (i.e., estrone and 17 β -estradiol [E2]).

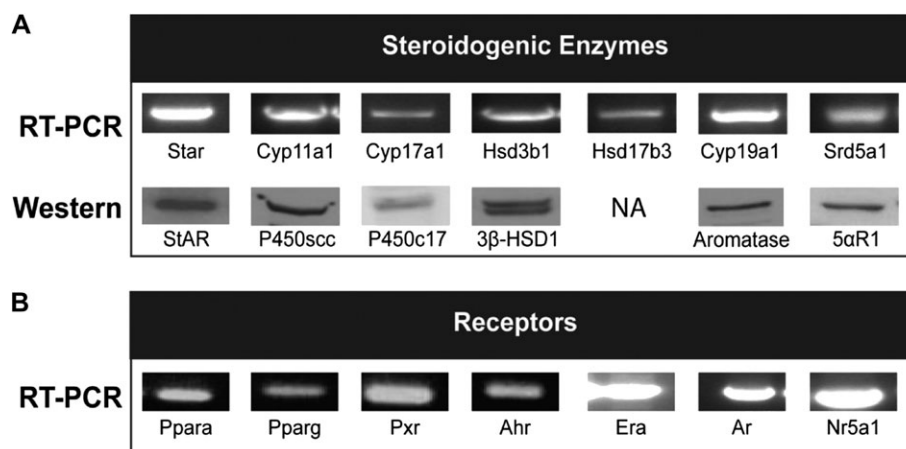


FIG. 2. Steroidogenic enzyme and receptor expression in BLTK1 cells. Expression of steroidogenic enzymes (A) and receptors (B) in untreated BLTK1 cells was evaluated by RT-PCR and Western blotting. mRNA and/or protein expression were detected for all steroidogenic enzymes. PPAR α , PPAR γ , pregnane X receptor, AhR, ER α , AR, and steroidogenic factor 1 expression were also confirmed. NA: antibody not available.

Steroidogenic enzyme messenger RNA (mRNA) and protein were detected in BLTK1 cells by RT-PCR and/or Western blotting, confirming the expression of all required steroidogenic enzymes (Fig. 2A). In addition, mRNA for several potential regulatory factors including LHCGR (data not shown), estrogen receptor (ER), androgen receptor (AR), and steroidogenic factor 1 (SF-1), peroxisome proliferator-activated receptors (PPAR α and PPAR γ), the pregnane X receptor (PXR), and the aryl hydrocarbon receptor (AhR) were also detected (Fig. 2B). However, mRNA for progesterone receptor, glucocorticoid receptor, or the liver receptor homolog 1 was not detected in BLTK1 cells despite verification of RT-PCR primer specificity and functionality in mouse Hepa1c1c7 cells (data not shown).

Induction of Steroidogenesis by FSK and rhCG

Temporal profiles of intracellular cAMP as well as P and T levels in media were evaluated in response to 3 ng/ml rhCG or 10 μ M FSK by EIA (Fig. 3). Intracellular cAMP was induced by FSK at 30 min ($\sim$ 120 pmol/ml, $\sim$ 10-fold) and by 1 h in response to rhCG (635 pmol/ml, 60-fold). However, levels quickly diminish such that no intracellular cAMP was detected by 8 h. Maximum P levels (200 ng/ml, eightfold) were observed at 2 h in response to rhCG and FSK, followed by a steady decline due to metabolism to androgens and estrogens. In contrast, T levels gradually increased reaching a maximum of $\sim$ 200 pg/ml (sevenfold) at 48 h, with significant increases detected as early as 1 h posttreatment.

Concentration-dependent induction of intracellular cAMP and secreted P and T was evaluated at 4 h when cAMP can still be detected (Figs. 4A–C). 17 β -Estradiol (E2) was evaluated at 48 h (Fig. 4D) as it was not consistently detected at 4 h (data not shown). cAMP, P, and T were induced 25-, 10-, and 4-fold, respectively, at 4 h, whereas E2 was induced $\sim$ fourfold by 48 h. The EC₅₀ for cAMP induction was $>$ 24 ng/ml for rhCG and $>$ 29 μ M for FSK. Meanwhile, EC₅₀ values of 1 ng/ml

rhCG and 9 μ M FSK were conserved for both P and T induction, whereas E2 EC₅₀ values were 10 ng/ml for rhCG and 9 μ M for FSK.

Intracellular cAMP levels are not only regulated by synthesis but also degradation, which is regulated by cyclic nucleotide phosphodiesterase enzymes (Chen *et al.*, 2007). The phosphodiesterase inhibitor IBMX maximizes cAMP levels in order to further induce steroidogenesis. However, IBMX cotreatment with rhCG or FSK did not increase T levels further, albeit rhCG and FSK potencies were greater (Fig. 5; T EC₅₀ with and without IBMX by FSK: 0.1 μ M vs. 9.4 μ M and rhCG: 0.1 ng/ml vs. 0.9 ng/ml, respectively).

rhCG- and FSK-Elicited Steroidogenic Gene Expression

The effect of rhCG and FSK on the gene expression of steroidogenic enzymes including steroidogenic acute regulatory protein (*Star*) was evaluated by QRT-PCR at 24 h. rhCG elicited concentration-dependent induction of *Star*, *Hsd3b6*, *Hsd17b3*, and *Srd5a1* mRNA levels (Fig. 6). In contrast, only *Star* and *Cyp17a1* mRNA exhibited concentration-dependent induction by FSK at 24 h. *Star*, the rate-limiting step in steroidogenesis, was induced $\sim$ 20-fold by rhCG and $\sim$ 10-fold by FSK. Temporal gene expression (Fig. 6) also revealed differences in the induction kinetics of *Star*, *Hsd3b6*, and *Cyp19a1* by rhCG and FSK. Most notably were differences in the induced levels of *Hsd17b3* and *Srd5a1*. These differences are likely a result of differences in pathway activation. FSK directly induces cAMP levels by the activating adenylyl cyclase, whereas rhCG induction via the LHCGR activates adenylyl cyclase and can also activate alternate signal transduction pathways such as the release of arachidonic acid and its metabolites (Ronco *et al.*, 2002; Wang and Stocco, 1999). Furthermore, *Cyp17a1* induction was only seen with FSK, while rhCG inhibited expression at 2 and 24 h. There were negligible effects on *Cyp11a1* and *Hsd3b1* in both concentration-

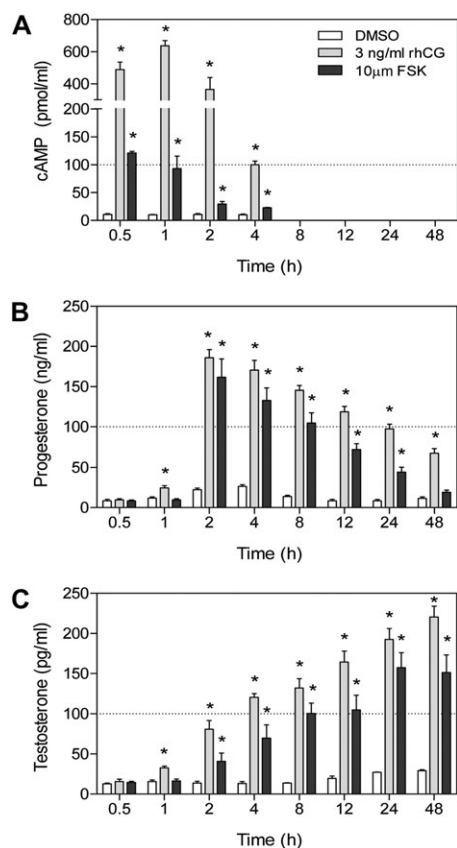


FIG. 3. Temporal rhCG- and FSK-induced cAMP, P and T levels. Intracellular cAMP (A) as well as P (B) and T (C) levels in media were evaluated in response to 3 ng/ml rhCG (light bars) or 10 μM FSK (dark bars) relative to DMSO vehicle controls (white bars). Both treatments elicited similar temporal profiles consisting of maximal cAMP levels reached early at 0.5–1 h followed by decrease and was undetectable by 8 h, whereas P reached maximum induction at 2 h followed by a gradual decrease over time. In contrast, T levels steadily increased throughout the time course. Data represent mean ± SE of three replicates, **p* < 0.05 vs. DMSO controls.

response and temporal evaluation (data not shown). More detailed studies are warranted to elucidate the mechanisms responsible for the differential effects of rhCG and FSK on steroidogenic gene expression.

Effects of Endocrine Disrupting Compounds on Testosterone Levels

The effect of selected EDCs, reported to elicit reproductive toxicity (Table 1), was assessed in BLTK1 cells by evaluating T levels. They include the triazine herbicide atrazine (Pogrmic-Majkic *et al.*, 2010; Trentacoste *et al.*, 2001), the broad spectrum herbicidal active ingredient glyphosate used in commercially available Roundup (Romano *et al.*, 2010; Walsh *et al.*, 2000), the conazole antifungal prochloraz (Kjaerstad *et al.*, 2010; Vinggaard *et al.*, 2005), the antifungal/antibacterial triclosan (Kumar *et al.*, 2009; Zorrilla *et al.*, 2009), and the diethylhexyl phthalate (DEHP) metabolite monoethylhexyl phthalate (MEHP)

(Fan *et al.*, 2010; Ge *et al.*, 2007; Gunnarsson *et al.*, 2008). None of these chemicals elicited significant cytotoxicity in the concentration range examined (Supplementary fig. S1). Compounds were evaluated for concentration-dependent effect on basal T levels and at a single cotreatment concentration to investigate potential interactions with rhCG-mediated stimulation of T levels (Fig. 7).

T levels were induced twofold (60 pg/ml) and 10-fold (250 pg/ml) by atrazine and MEHP, respectively. No other examined chemical altered basal T levels. However, steroidogenesis in Leydig cells is stimulated by LH via the LHCGR during late gestation first measurable at gestational day (GD) 15 and peaking at GD 18 in the mouse (O'Shaughnessy *et al.*, 2006), and throughout gestation in humans, with a similar programming window for androgen-dependent masculinization extrapolated between humans and rodents (Welsh *et al.*, 2008). LH stimulation is also crucial during puberty and adulthood to stimulate T biosynthesis to support Sertoli cell spermatogenesis. Consequently, cotreatment studies with 3 ng/ml rhCG were conducted to mimic Leydig cell stimulation. Atrazine (300 μM) inhibited rhCG-induced T levels ~30% (rhCG alone ~100 pg/ml T vs. Atrazine + rhCG ~70 pg/ml T). Similarly, 100 μM MEHP inhibited rhCG-elicited T induction ~20% (rhCG alone ~100 pg/ml vs. MEHP cotreatment ~80 pg/ml). In contrast, prochloraz did not significantly alter basal T levels but cotreatment decreased rhCG-induced T to control levels (rhCG ~100 pg/ml vs. prochloraz + rhCG ~50 pg/ml vs. DMSO ~40 pg/ml). Similarly, 30 μM triclosan had no effect, yet cotreatment decreased rhCG-elicited T induction ~25% (rhCG alone ~100 pg/ml vs. triclosan + rhCG cotreatment ~75 pg/ml). Glyphosate, 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD), and RDX did not induce or alter rhCG induction of T.

DISCUSSION

The *in vitro* evaluation of steroidogenesis is important for screening potential reproductive and developmental toxicants (Sanderson, 2006; Whitehead and Rice, 2006). Current models include human H295R adrenocarcinoma cells and several rodent Leydig cell lines including MA-10, mLTC, and R2C. Human H295R cells (American Type Culture Collection CLR-2128) were isolated from a pluripotent adrenocortical carcinoma in a 48-year-old black female, produce detectable basal aldosterone, cortisol, androgens, and E2 (Gracia *et al.*, 2006), and are part of the Endocrine Disruptor Screening Program to screen for effects on T and E2 biosynthesis (USEPA, 2009). However, H295R cells do not express functional adrenocorticotrophic hormone receptors (a member of the cell surface 7-transmembrane domain superfamily of G-protein-coupled receptors), and therefore, steroidogenesis must be stimulated using FSK. Moreover, besides glucocorticoids and mineralocorticoids, the adrenal cortex normally only produces dehydroepiandrosterone sulfate and androstenedione (precursors of T) and does not

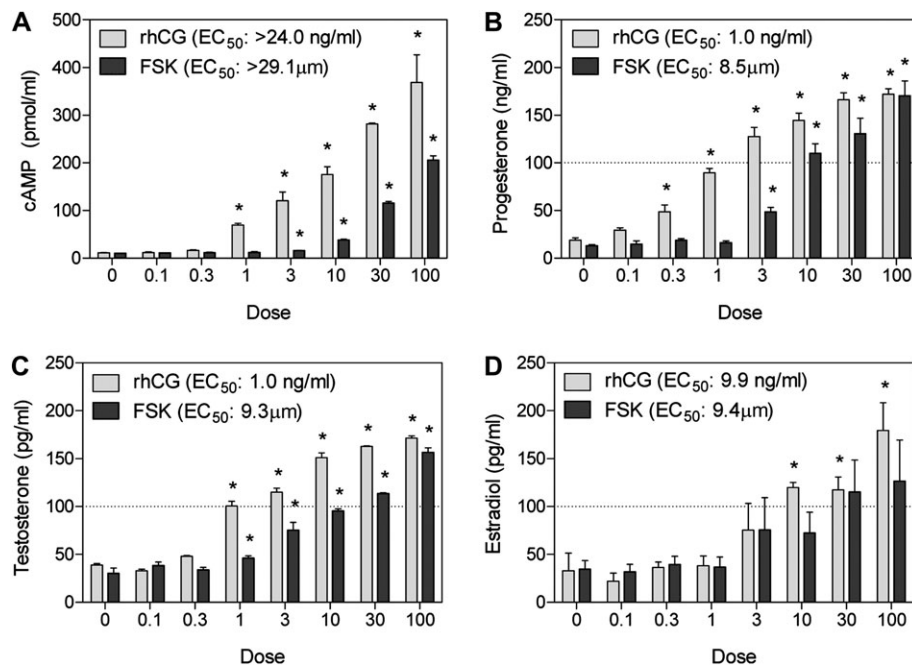


FIG. 4. Concentration-dependent induction of intracellular cAMP and secreted P, T, and E2 by rhCG and FSK. Intracellular cAMP concentrations (A) and secreted P (B) and T (C) levels in media induced by rhCG (ng/ml; light bars) and FSK (μ M; dark bars) at 4 h. E2 (D) levels in media were measured 48 h after treatment. EC_{50} values were calculated by the ToxResponse modeler. Data represent mean $\pm$ SE of three replicates, * p < 0.05 versus DMSO controls.

express the 17-beta-hydroxysteroid dehydrogenase necessary for T biosynthesis (Hinson and Raven, 1996). As such, it has been suggested that H295R cells have the characteristics of zonally

undifferentiated human fetal adrenal cells (Gracia *et al.*, 2006). Only Leydig cells synthesize T and E2 directly from cholesterol (Scott *et al.*, 2009). In ovarian steroidogenesis, theca and

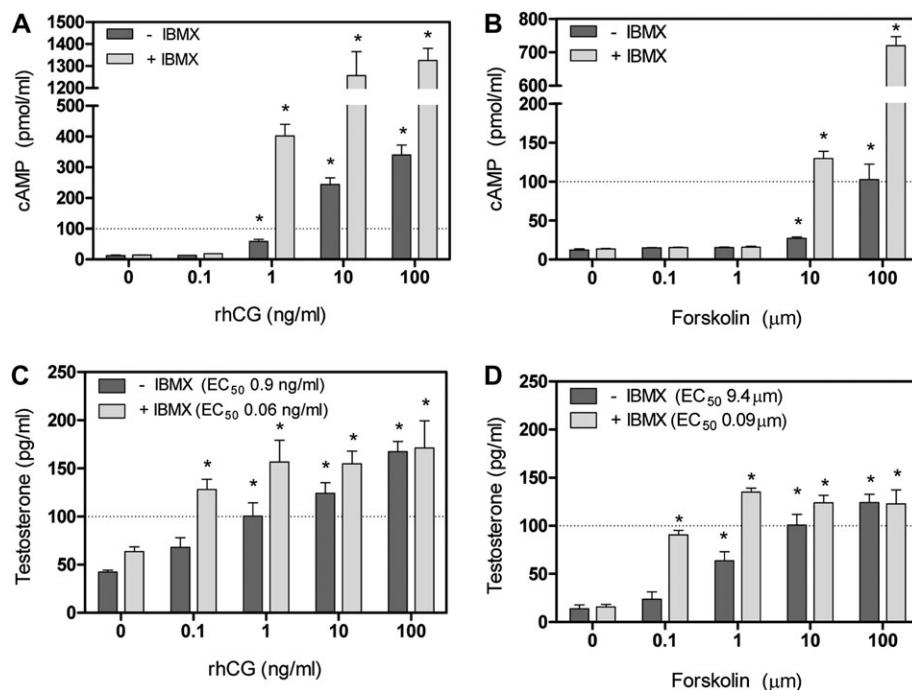


FIG. 5. Effect of the phosphodiesterase inhibitor IBMX on rhCG- and FSK-elicited induction of T. Concentration-dependent induction of cAMP (A and B) and T (C and D) by rhCG (0.1–100 ng/ml) or FSK (0.1–100 μ M) alone (dark bars) or in the presence of 0.2mM IBMX (light bars) for 4 h. Although IBMX did not affect T induction, cAMP levels were significantly increased, and the potency of rhCG and FSK were enhanced. Data represent mean $\pm$ SE of three independent replicates, * p < 0.05 versus DMSO controls.

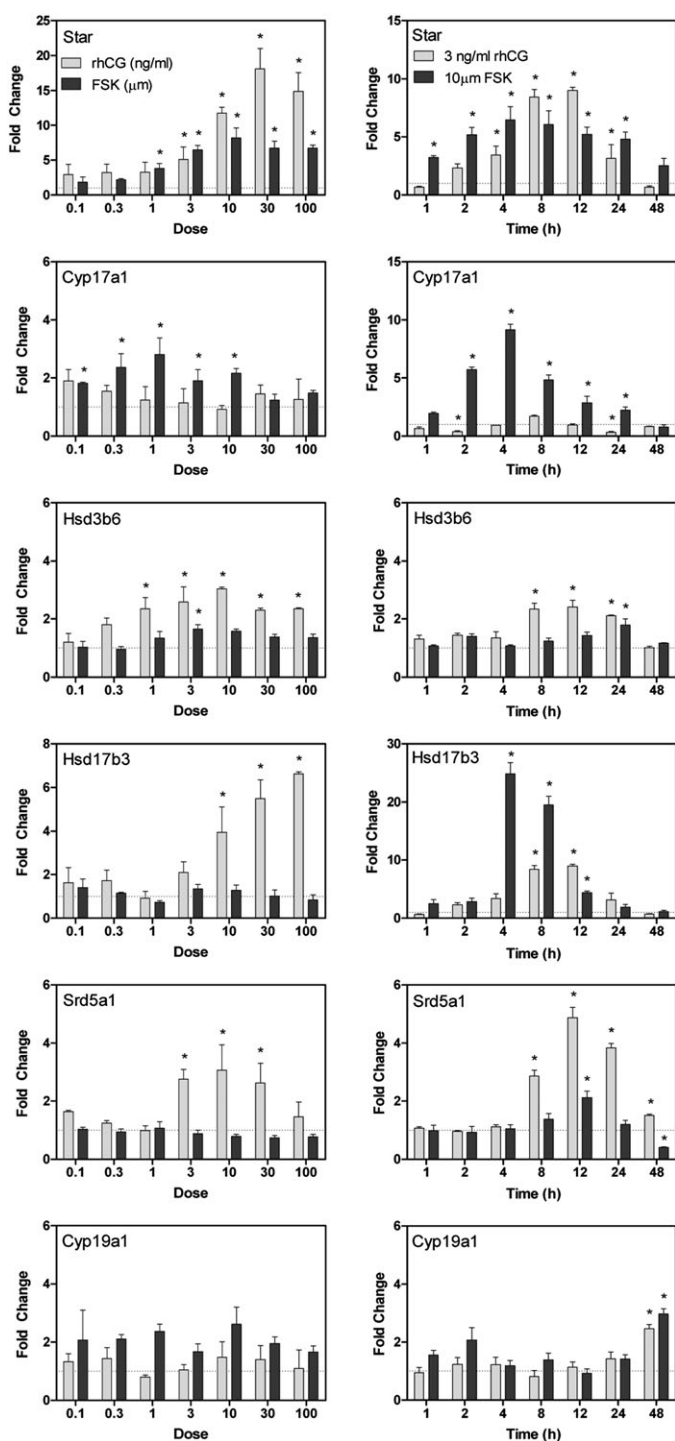


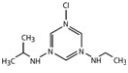
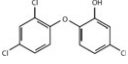
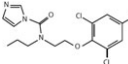
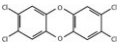
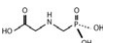
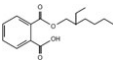
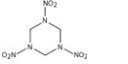
FIG. 6. Concentration- and time-dependent induction of steroidogenic gene expression. The concentration-dependent (24 h, left column) and temporal (right column) effects of rhCG (ng/ml, light bars) and FSK (μ M, dark bars) on *Star*, *Cyp17a1*, *Hsd3b6*, *Hsd17b3*, *Srd5a1*, and *Cyp19a1* mRNA levels were examined using QRT-PCR. *Cyp11a1* mRNA expression was not affected by treatment (data not shown). Data represent mean $\pm$ SE of three replicates, * p < 0.05 versus DMSO controls.

granulosa cells produce P, while only theca cells metabolize P to T, and subsequent aromatization to estrogens or 5α reduction to DHT occurs in granulosa cells (Havelock *et al.*, 2004). Steroidogenic theca cell line development has not been successful, and as such, rodent models are of Leydig cell origin such as mouse MA-10 and mLTC-1 models as well as rat R2C cells. MA-10 and mLTC1 cells were established from transplantable M548OP Leydig tumors carried in C57Bl/6 mice that exhibit P induction by rhCG but lacked the enzymes required for subsequent metabolism to T (Ascoli, 1981; Freeman and Ascoli, 1981; Rebois, 1982). However, recent studies report mLTC-1 cells can produce low levels of androgens, with 100 IU/ml hCG eliciting a maximum of 600 pmol/well P and 10 pmol/well T in a similar 24-well plating format as used herein (Panesar *et al.*, 2003). These levels are lower than those observed in resting unstimulated BLTK1 cells. Rat R2C cells lack functional LHCGR expression and are not rhCG inducible but exhibit high basal steroidogenesis activity (Stocco and Chen, 1991).

In contrast, BLTK1 cells produce progestogens, androgens, and E2 in response to trophic hormone stimulus via LHCGR, the 7-transmembrane G-protein-coupled cell surface receptor, similar to T biosynthesis *in vivo* following LH secretion from the anterior pituitary (Chen *et al.*, 2007). Although human steroidogenesis prefers the $\Delta 5$ pathway over the $\Delta 4$ pathway in mice, the enzymatic steps are highly conserved facilitating extrapolation of murine BLTK1 results to humans (Scott *et al.*, 2009). Gestation is the critical period of exposure to EDCs that targets fetal Leydig cells causing reproductive and development toxicity. Fetal Leydig cells are observed as early as GD 12.5 in mice or week 6 in humans (O'Shaughnessy *et al.*, 2006; Rahman and Huhtaniemi, 2004). Murine fetal Leydig cells do not express measurable LHCGR until GD 15.5, and murine fetal T production is LH independent (Yao and Barsoum, 2007). However, human fetal Leydig cells do express LHCGR and are dependent on LH and hCG stimulation of steroidogenesis for masculinization during development (O'Shaughnessy *et al.*, 2006). Though BLTK1 cells are not murine fetal Leydig cells, they retain rhCG-inducible steroidogenesis characteristic of murine adult Leydig cells and both fetal and adult human Leydig cells. The ability to evaluate rhCG-inducible steroidogenesis provides a unique functional feature that is important for screening EDC interactions that may occur during human fetal and adult exposure.

During gestation, LH stimulates Leydig cell proliferation and T production, whereas during puberty and adulthood, LH stimulates T biosynthesis to indirectly support spermatogenesis by contributing to Sertoli cell stimulation. rhCG also promotes cell proliferation or apoptosis, but no such effects were observed on BLTK1 cells (data not shown). rhCG is used clinically as an LH substitute to treat LH deficiency and in research to stimulate steroidogenesis (Hakola *et al.*, 1998), due to its longer half-life (~ 24 h compared with ~ 4 h for LH) (Yen *et al.*, 1968). Like LH, rhCG induces steroidogenesis via the LHCGR and initiates the same signaling pathways resulting in

TABLE 1
Reported Evaluated Endocrine Disrupting Compounds (EDCs) That Alter T Levels

Common name	Chemical name	CAS no.	Molecular weight (g/mol)	Structure	Reproductive toxicity
Atrazine	2-chloro-4-ethylamino-6-isopropylamino-s-triazine	1912-24-9	215.68		Decreased serum T, dysregulation of steroidogenesis (Pogrmic-Majkic <i>et al.</i> , 2010; Trentacoste <i>et al.</i> , 2001)
Triclosan	2,4,4'-trichloro-2'-hydroxydiphenyl ether	3380-34-5	289.54		Antiandrogenic, decreased T production by Leydig cells, decreased serum T levels (Kumar <i>et al.</i> , 2009; Zorrilla <i>et al.</i> , 2009)
Prochloraz	N-propyl-N-(2,4,6-trichlorophenoxy) ethyl-imidazole-1-carboxamide	67747-09-5	376.67		Decreased serum T levels, AR antagonism, altered androgen-dependent gene expression (Kjaerstad <i>et al.</i> , 2010; Vinggaard <i>et al.</i> , 2005)
Dioxin (TCDD)	2,3,7,8-tetrachlorodibenzo-p-dioxin	1746-01-6	321.97		Decreased intratesticular T, inhibition of steroidogenic gene expression (Adamsson <i>et al.</i> , 2009; Fukuzawa <i>et al.</i> , 2004; Lai <i>et al.</i> , 2005)
Glyphosate	N-(phosphonomethyl) glycine	1071-83-6	169.07		Inhibition of steroidogenic enzymes expression, decreased T production (Romano <i>et al.</i> , 2010; Walsh <i>et al.</i> , 2000)
MEHP	2-(2-ethylhexoxycarbonyl) benzoic acid	4376-20-9	278.35		Decreased serum T levels, Leydig cell hyperplasia (Ge <i>et al.</i> , 2007; Gunnarsson <i>et al.</i> , 2008)
RDX	1,3,5-trinitroperhydro-1,3,5-triazine	121-82-4	222.12		Altered reproductive capacity (Mukhi and Patino, 2008; Zhang <i>et al.</i> , 2006, 2008)

a robust *in vivo* and *in vitro* steroidogenic response (Dufau *et al.*, 1984; Hakola *et al.*, 1998; Payne and Youngblood, 1995). Interestingly, LH/rhCG levels required for maximum

cAMP synthesis are 15-fold higher than the level needed for maximal T production (Chen *et al.*, 2007), suggesting that minimal changes in cAMP can affect steroidogenesis. Our

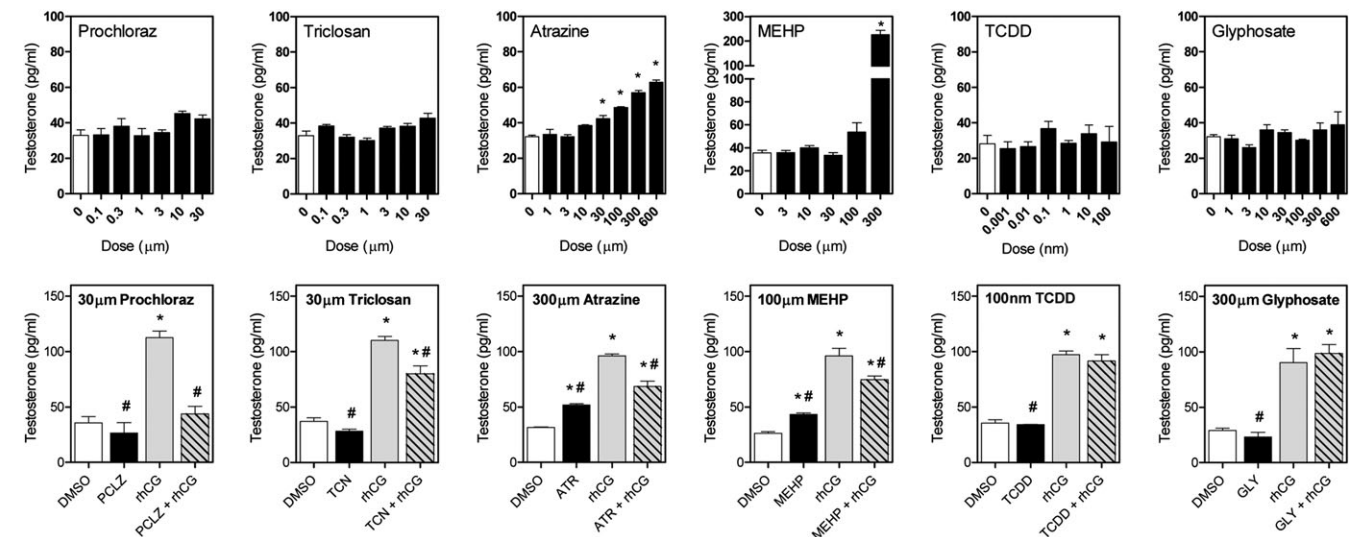


FIG. 7. Effects of EDCs on T levels. The concentration-dependent effect of prochloraz, atrazine, glyphosate, triclosan, MEHP, and TCDD on T levels were evaluated after 4 h (top row, black bars) relative to DMSO controls (white bars). The effect of 30µM prochloraz, 300µM atrazine, 300µM glyphosate, 30µM triclosan, 100µM MEHP, and 100nM TCDD on rhCG-induced (3 ng/ml rhCG) T levels 4 h after treatment (bottom row). RDX had no effect on basal or rhCG-induced T levels (data not shown). Data represent mean ± SE of three replicates, **p* < 0.05 versus DMSO controls, #*p* < 0.05 versus rhCG.

results demonstrating that IBMX cotreatment increased rhCG potency is consistent with the potentiation of cAMP induction with no effect on maximum T induction, suggesting maximum steroidogenic capacity has been achieved. It also highlights the important role of phosphodiesterases in the regulation of steroidogenesis. For example, Leydig cells isolated from phosphodiesterase 8A (PDE8A) null mice are more sensitive to LH induction but induced T levels were comparable in wild-type and PDE8A null mice (Vasta *et al.*, 2006). These data are consistent with IBMX-treated BLTK1 cells and the enhanced potency of rhCG and FSK but comparable induced T levels.

The acute effect of LH/rhCG is activation of cAMP signaling that induces rapid mobilization of cholesterol stores and induction of steroidogenic enzyme activity. However, long-term effects include the induction of steroidogenic gene expression (Chen *et al.*, 2007). Differential gene expression is necessary as increased steroidogenesis in response to trophic hormone requires *de novo* protein synthesis, such as StAR induction (Stocco and Chen, 1991). Consequently, in addition to effects on T biosynthesis, comprehensive screening should also include effects on cAMP as well as changes in steroidogenic gene expression.

BLTK1 cells express all the necessary enzymes required for T biosynthesis and metabolism, as well as LHCG and several other regulators of steroidogenesis. rhCG and FSK, a direct activator of adenylyl cyclase, induce steroidogenesis in BLTK1 murine Leydig cells as indicated by time-dependent increases in cAMP, P, T, and E2 levels and steroidogenic gene expression. Robust induction of intracellular cAMP occurred rapidly after rhCG and FSK treatment. Overall, rhCG and FSK elicited comparable induction of intracellular cAMP, steroidogenic enzyme mRNA expression, and altered P, T, and E2 levels (with and without IBMX). Furthermore, BLTK1 cells exhibited robust and consistent induction of steroidogenesis for up to 30 passages (data not shown), whereas H295R cell assays are limited to passages 5–10 (Hecker *et al.*, 2006; USEPA, 2009).

Intracellular cAMP, as well as P, T, and E2 levels in media, exhibited concentration-dependent induction following rhCG and FSK treatment in the absence and presence of IBMX. Selected EDCs also altered steroidogenesis in BLTK1 cells consistent with reports in other models. For example, prochloraz, a residential and commercial imidazole fungicide that inhibits C17-20 lyase (Cyp17a1) and aromatase (Cyp19a1), as well as antagonizes AR and ER activity (Kjaerstad *et al.*, 2010; Ohlsson *et al.*, 2009; Vinggaard *et al.*, 2006), completely blocked T induction by rhCG. However, prochloraz alone did not alter basal T levels most likely due to the low basal level of production in BLTK1 cells. In contrast, prochloraz reduced basal T levels in H295R adrenal cells, which have relatively high basal T levels (Hecker *et al.*, 2006). Similarly, atrazine, which decreases circulating T levels and antagonizes rhCG-induced T production in primary Leydig cells (Friedmann, 2002; Pogrmic *et al.*, 2009; Pogrmic-

Majkic *et al.*, 2010; Stoker *et al.*, 2000; Victor-Costa *et al.*, 2010), inhibited rhCG-induced T levels ~30% in BLTK1 cells. However, atrazine alone induced modest increases in T levels, consistent with results in rat primary Leydig cell cultures (Pogrmic-Majkic *et al.*, 2010). This modest induction of basal T is consistent with atrazine acting as a phosphodiesterase inhibitor (Roberge *et al.*, 2004). Triclosan did not alter basal levels when administered alone but decreased rhCG-induced T levels in cotreatment studies in agreement with its reported antiandrogenic activity (Kumar *et al.*, 2008, 2009; Zorrilla *et al.*, 2009). Despite the confirmed expression of the AhR in BLTK1 cells, and previous studies *in vivo* suggesting AhR-mediated inhibition of steroidogenic expression and reduction of fetal intratesticular T levels (Adamsson *et al.*, 2009; Fukuzawa *et al.*, 2004; Lai *et al.*, 2005), TCDD had no effect on T levels in BLTK1 cells. A recent study in mouse primary Leydig cells has also demonstrated that TCDD does not alter Leydig cell steroidogenesis (Naville *et al.*, 2011). Collectively, these differences in agonist and antagonist profiles suggest EDC-elicited effects on steroidogenesis use various modes of action that warrant further investigation. Moreover, these results highlight the need for more comprehensive screens that not only monitor effects on T and E2 levels, but also examine effects on steroidogenic enzyme expression and activity, or even alternative signaling cascades in order to elucidate EDC-specific adverse outcome pathways that are relevant to humans and wildlife.

BLTK1 cells can also be used to evaluate the concentration-dependent effects of specific metabolites. The mono-ester metabolite, MEHP, is thought to be responsible for DEHP-elicited *in vivo* reproductive and developmental toxicity in rodents at high doses. MEHP increased T levels in BLTK1 cells, consistent with results obtained in mouse mLTC-1 Leydig cells (Gunnarsson *et al.*, 2008). Furthermore, the inhibition of rhCG-induced T levels by MEHP is also reported in mouse MA-10 Leydig cell line, consistent with the proposed antagonistic effect of phthalates on T levels (Clewett *et al.*, 2010; Fan *et al.*, 2010). Although RDX elicits reproductive and developmental effects in zebrafish, earthworms, and crickets, no effects on T levels were observed in BLTK1 cells, suggesting RDX may not elicit steroidogenic effects in mammalian systems or that other indirect interactions may be involved (Zhang *et al.*, 2006, 2008). Glyphosate also had no effect on T levels in BLTK1 cells, despite previous studies showing effects on steroidogenesis in rat and mouse models (Romano *et al.*, 2010; Walsh *et al.*, 2000). However, these studies evaluated Roundup formulations, which are complex mixtures with glyphosate as the active pesticide ingredient. Glyphosate alone only has marginal effects compared with the Roundup formulation (Walsh *et al.*, 2000), suggesting formulations as well as active and inert components should be examined. Nevertheless, more comprehensive temporal and concentration-response follow-up studies are required that include assessing cAMP, P, T, and E2 levels and gene

expression prior to any conclusions regarding the steroidogenic effects of these chemicals.

In summary, current protocols and models are inadequate to screen the universe of chemicals, metabolites, and mixtures that may alter steroidogenesis (Sanderson, 2006; Whitehead and Rice, 2006; Wilson *et al.*, 2008). BLTK1 cells are a novel complementary rhCG-inducible Leydig-based model that can be used to assess effects on steroidogenic gene expression, intracellular cAMP, and P, T, and E2 levels in media. Their consistent response characteristics and inducibility over 30 passages also make this cell line attractive for high-throughput screening. Comprehensive characterization of effects on intermediate steroid biosynthesis, including pregnenolone, 17-hydroxyprogesterone, DHEA, androstenedione, estrone, and DHT, as well as the differential expression of steroidogenic enzymes will also facilitate the elucidation of modes of action relevant to adverse outcome pathways in humans and other relevant species.

SUPPLEMENTARY DATA

Supplementary data are available online at <http://toxsci.oxfordjournals.org/>.

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REFERENCES

- Adamsson, A., Simanainen, U., Viluksela, M., Paranko, J., and Toppari, J. (2009). The effects of 2,3,7,8-tetrachlorodibenzo-p-dioxin on foetal male rat steroidogenesis. *Int. J. Androl.* **32**, 575–585.
- Andersen, M. E., and Krewski, D. (2009). Toxicity testing in the 21st century: Bringing the vision to life. *Toxicol. Sci.* **107**, 324–330.
- Ascoli, M. (1981). Characterization of several clonal lines of cultured Leydig tumor cells: Gonadotropin receptors and steroidogenic responses. *Endocrinology* **108**, 88–95.
- Borgert, C. J., Mihaich, E. M., Quill, T. F., Marty, M. S., Levine, S. L., and Becker, R. A. (2011). Evaluation of EPA's Tier 1 Endocrine Screening Battery and recommendations for improving the interpretation of screening results. *Regul. Toxicol. Pharmacol.* **59**, 397–411.
- Breithaupt, H. (2006). The costs of REACH. REACH is largely welcomed, but the requirement to test existing chemicals for adverse effects is not good news for all. *EMBO Rep.* **7**, 968–971.
- Brown, V. J. (2003). REACHing for chemical safety. *Environ. Health Perspect.* **111**, A766–A769.
- Burgoon, L. D., and Zacharewski, T. R. (2008). Automated quantitative dose-response modeling and point of departure determination for large toxicogenomic and high-throughput screening data sets. *Toxicol. Sci.* **104**, 412–418.
- Chen, H., Midzak, A., Luo, L., and Zirkin, B. R. (2007). In *Aging and the Decline of Androgen Production* (A. H. Payne and M. P. Hardy, Eds.). Humana Press Inc., Totowa, NJ.
- Clewell, R. A., Campbell, J. L., Ross, S. M., Gaido, K. W., Clewell, H. J., III, and Andersen, M. E. (2010). Assessing the relevance of in vitro measures of phthalate inhibition of steroidogenesis for in vivo response. *Toxicol. In Vitro* **24**, 327–334.
- Cooper, R. L. (2009). Current developments in reproductive toxicity testing of pesticides. *Reprod. Toxicol.* **28**, 180–187.
- Daston, G. P. (2007). Genomics and developmental risk assessment. *Birth Defects Res.* **79**, 1–7.
- Daston, G. P., Chapin, R. E., Scialli, A. R., Piersma, A. H., Carney, E. W., Rogers, J. M., and Friedman, J. M. (2010). A different approach to validating screening assays for developmental toxicity. *Birth Defects Res. B Dev. Reprod. Toxicol.* **89**, 526–530.
- Dix, D. J., Houck, K. A., Martin, M. T., Richard, A. M., Setzer, R. W., and Kavlock, R. J. (2007). The ToxCast program for prioritizing toxicity testing of environmental chemicals. *Toxicol. Sci.* **95**, 5–12.
- Dufau, M. L., Winters, C. A., Hattori, M., Aquilano, D., Baranao, J. L., Nozu, K., Baukal, A., and Catt, K. J. (1984). Hormonal regulation of androgen production by the Leydig cell. *J. Steroid Biochem.* **20**, 161–173.
- Edwards, T. M., Moore, B. C., and Guillette, L. J., Jr (2006). Reproductive dysgenesis in wildlife: A comparative view. *Int. J. Androl.* **29**, 109–121.
- Fan, J., Traore, K., Li, W., Amri, H., Huang, H., Wu, C., Chen, H., Zirkin, B., and Papadopoulos, V. (2010). Molecular mechanisms mediating the effect of mono-(2-ethylhexyl) phthalate on hormone-stimulated steroidogenesis in MA-10 mouse tumor Leydig cells. *Endocrinology* **151**, 3348–3362.
- Freeman, D. A., and Ascoli, M. (1981). Desensitization to gonadotropins in cultured Leydig tumor cells involves loss of gonadotropin receptors and decreased capacity for steroidogenesis. *Proc. Natl. Acad. Sci. U.S.A.* **78**, 6309–6313.
- Friedmann, A. S. (2002). Atrazine inhibition of testosterone production in rat males following peripubertal exposure. *Reprod. Toxicol.* **16**, 275–279.
- Fukuzawa, N. H., Ohsako, S., Wu, Q., Sakaue, M., Fujii-Kuriyama, Y., Baba, T., and Tohyama, C. (2004). Testicular cytochrome P450scc and LHR as possible targets of 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) in the mouse. *Mol. Cell. Endocrinol.* **221**, 87–96.
- Ge, R. S., Chen, G. R., Tanrikut, C., and Hardy, M. P. (2007). Phthalate ester toxicity in Leydig cells: Developmental timing and dosage considerations. *Reprod. Toxicol.* **23**, 366–373.
- Gracia, T., Hilscherova, K., Jones, P. D., Newsted, J. L., Zhang, X., Hecker, M., Higley, E. B., Sanderson, J. T., Yu, R. M., Wu, R. S., *et al.* (2006). The H295R system for evaluation of endocrine-disrupting effects. *Ecotoxicol. Environ. Saf.* **65**, 293–305.
- Gray, L. E., Jr, Wilson, V. S., Stoker, T., Lambright, C., Furr, J., Noriega, N., Howdeshell, K., Ankley, G. T., and Guillette, L. (2006). Adverse effects of environmental antiandrogens and androgens on reproductive development in mammals. *Int. J. Androl.* **29**, 96–104; discussion 105–108.
- Gunnarsson, D., Leffler, P., Ekwurtzel, E., Martinsson, G., Liu, K., and Selstam, G. (2008). Mono-(2-ethylhexyl) phthalate stimulates basal steroidogenesis by a cAMP-independent mechanism in mouse gonadal cells of both sexes. *Reproduction* **135**, 693–703.
- Hakola, K., Haavisto, A. M., Pierroz, D. D., Aebi, A., Rannikko, A., Kirjavainen, T., Aubert, M. L., and Huhtaniemi, I. (1998). Recombinant forms of rat and human luteinizing hormone and follicle-stimulating hormone; comparison of functions in vitro and in vivo. *J. Endocrinol.* **158**, 441–448.
- Harding, A. K., Daston, G. P., Boyd, G. R., Lucier, G. W., Safe, S. H., Stewart, J., Tillitt, D. E., and Van Der Kraak, G. (2006). Endocrine disrupting chemicals research program of the U.S. Environmental Protection Agency: Summary of a peer-review report. *Environ. Health Perspect.* **114**, 1276–1282.

- Hartung, T., and Rovida, C. (2009). Chemical regulators have overreached. *Nature* **460**, 1080–1081.
- Havelock, J. C., Rainey, W. E., and Carr, B. R. (2004). Ovarian granulosa cell lines. *Mol. Cell. Endocrinol.* **228**, 67–78.
- Hecker, M., Newsted, J. L., Murphy, M. B., Higley, E. B., Jones, P. D., Wu, R., and Giesy, J. P. (2006). Human adrenocarcinoma (H295R) cells for rapid in vitro determination of effects on steroidogenesis: Hormone production. *Toxicol. Appl. Pharmacol.* **217**, 114–124.
- Higley, E. B., Newsted, J. L., Zhang, X., Giesy, J. P., and Hecker, M. (2010). Assessment of chemical effects on aromatase activity using the H295R cell line. *Environ. Sci. Pollut. Res. Int.* **17**, 1137–1148.
- Hinson, J. P., and Raven, P. W. (1996). Adrenal morphology and hormone synthesis and regulation. In *The Adrenal in Toxicology: Target Organ and Modulator of Toxicity* (P. W. Harvey, Ed.), pp. 23–52. Taylor & Francis Ltd, Bristol, PA.
- Hofer, T., Gerner, I., Gundert-Remy, U., Liebsch, M., Schulte, A., Spielmann, H., Vogel, R., and Wettig, K. (2004). Animal testing and alternative approaches for the human health risk assessment under the proposed new European chemicals regulation. *Arch. Toxicol.* **78**, 549–564.
- Holsapple, M. P., Afshari, C. A., and Lehman-McKeeman, L. D. (2009). Forum Series: The “Vision” for Toxicity Testing in the 21st Century: Promises and conundrums. *Toxicol. Sci.* **107**, 307–308.
- Kananen, K., Markkula, M., el-Hefnawy, T., Zhang, F. P., Pauku, T., Su, J. G., Hsueh, A. J., and Huhtaniemi, I. (1996). The mouse inhibin alpha-subunit promoter directs SV40 T-antigen to Leydig cells in transgenic mice. *Mol. Cell. Endocrinol.* **119**, 135–146.
- Kimmel, C. A., Generoso, W. M., Thomas, R. D., and Bakshi, K. S. (1993). A new frontier in understanding the mechanisms of developmental abnormalities. *Toxicol. Appl. Pharmacol.* **119**, 159–165.
- Kjaerstad, M. B., Taxvig, C., Nellemann, C., Vinggaard, A. M., and Andersen, H. R. (2010). Endocrine disrupting effects in vitro of conazole antifungals used as pesticides and pharmaceuticals. *Reprod. Toxicol.* **30**, 573–582.
- Knudsen, T. B., and Daston, G. P. (2010). Virtual tissues and developmental systems biology. In *Comprehensive Toxicology*, 2nd ed. Chapter 23 (T. Knudsen and G. Daston, Eds.), Vol. 12, pp. 347–358. Elsevier Ltd, Oxford, U.K.
- Kumar, V., Balomajumder, C., and Roy, P. (2008). Disruption of LH-induced testosterone biosynthesis in testicular Leydig cells by triclosan: Probable mechanism of action. *Toxicology* **250**, 124–131.
- Kumar, V., Chakraborty, A., Kural, M. R., and Roy, P. (2009). Alteration of testicular steroidogenesis and histopathology of reproductive system in male rats treated with triclosan. *Reprod. Toxicol.* **27**, 177–185.
- Lai, K. P., Wong, M. H., and Wong, C. K. (2005). Inhibition of CYP450scc expression in dioxin-exposed rat Leydig cells. *J. Endocrinol.* **185**, 519–527.
- Marty, M. S., Allen, B., Chapin, R. E., Cooper, R., Daston, G. P., Flaws, J. A., Foster, P. M., Makris, S. L., Mylchreest, E., Sandler, D., et al. (2009). Inter-laboratory control data for reproductive endpoints required in the OPPTS 870.3800/OECD 416 reproduction and fertility test. *Birth Defects Res. B Dev. Reprod. Toxicol.* **86**, 470–489.
- Mukhi, S., and Patino, R. (2008). Effects of hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) in zebrafish: General and reproductive toxicity. *Chemosphere* **72**, 726–732.
- National Research Council. (2007). *Toxicity Testing in the 21st Century: A Vision and a Strategy*. National Academies Press, Washington, DC.
- Naville, D., Rebouret, D., Chauvin, M. A., Vega, N., Jalabert, A., Vigier, M., Loizon, E., Begeot, M., and Le Magueresse-Battistoni, B. (2011). Direct and indirect impact of 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) on adult mouse Leydig cells: An in vitro study. *Toxicol. Lett.* **207**, 251–257.
- Ohlsson, A., Ulleras, E., and Oskarsson, A. (2009). A biphasic effect of the fungicide prochloraz on aldosterone, but not cortisol, secretion in human adrenal H295R cells—Underlying mechanisms. *Toxicol. Lett.* **191**, 174–180.
- O’Shaughnessy, P. J., Baker, P. J., and Johnston, H. (2006). The foetal Leydig cell—Differentiation, function and regulation. *Int. J. Androl.* **29**, 90–95; discussion 105–108.
- Panesar, N. S., Chan, K. W., and Ho, C. S. (2003). Mouse Leydig tumor cells produce C-19 steroids, including testosterone. *Steroids* **68**, 245–251.
- Payne, A. H., and Youngblood, G. L. (1995). Regulation of expression of steroidogenic enzymes in Leydig cells. *Biol. Reprod.* **52**, 217–225.
- Pogrmic, K., Fa, S., Dakic, V., Kaisarevic, S., and Kovacevic, R. (2009). Atrazine oral exposure of peripubertal male rats down-regulates steroidogenesis gene expression in Leydig cells. *Toxicol. Sci.* **111**, 189–197.
- Pogrmic-Majkic, K., Fa, S., Dakic, V., Kaisarevic, S., and Kovacevic, R. (2010). Upregulation of peripubertal rat Leydig cell steroidogenesis following 24 h in vitro and in vivo exposure to atrazine. *Toxicol. Sci.* **118**, 52–60.
- Rahman, N. A., and Huhtaniemi, I. T. (2004). Testicular cell lines. *Mol. Cell. Endocrinol.* **228**, 53–65.
- Rebois, R. V. (1982). Establishment of gonadotropin-responsive murine Leydig tumor cell line. *J. Cell Biol.* **94**, 70–76.
- Roberge, M., Hakk, H., and Larsen, G. (2004). Atrazine is a competitive inhibitor of phosphodiesterase but does not affect the estrogen receptor. *Toxicol. Lett.* **154**, 61–68.
- Romano, R. M., Romano, M. A., Bernardi, M. M., Furtado, P. V., and Oliveira, C. A. (2010). Prepubertal exposure to commercial formulation of the herbicide glyphosate alters testosterone levels and testicular morphology. *Arch. Toxicol.* **84**, 309–317.
- Ronco, A. M., Moraga, P. F., and Llanos, M. N. (2002). Arachidonic acid release from rat Leydig cells: The involvement of G protein, phospholipase A2 and regulation of cAMP production. *J. Endocrinol.* **172**, 95–104.
- Sanderson, J. T. (2006). The steroid hormone biosynthesis pathway as a target for endocrine-disrupting chemicals. *Toxicol. Sci.* **94**, 3–21.
- Schenk, B., Weimer, M., Bremer, S., van der Burg, B., Cortvriendt, R., Freyberger, A., Lazzari, G., Pellizzer, C., Piersma, A., Schafer, W. R., et al. (2010). The ReProTect Feasibility Study, a novel comprehensive in vitro approach to detect reproductive toxicants. *Reprod. Toxicol.* **30**, 200–218.
- Scott, H. M., Mason, J. I., and Sharpe, R. M. (2009). Steroidogenesis in the fetal testis and its susceptibility to disruption by exogenous compounds. *Endocr. Rev.* **30**, 883–925.
- Sharpe, R. M. (2006). Pathways of endocrine disruption during male sexual differentiation and masculinization. *Best Pract. Res. Clin. Endocrinol. Metab.* **20**, 91–110.
- Shukla, S. J., Huang, R., Austin, C. P., and Xia, M. (2010). The future of toxicity testing: A focus on in vitro methods using a quantitative high-throughput screening platform. *Drug Discov. Today* **15**, 997–1007.
- Skakkebaek, N. E., Rajpert-De Meyts, E., and Main, K. M. (2001). Testicular dysgenesis syndrome: An increasingly common developmental disorder with environmental aspects. *Hum. Reprod.* **16**, 972–978.
- Stocco, D. M., and Chen, W. (1991). Presence of identical mitochondrial proteins in unstimulated constitutive steroid-producing R2C rat Leydig tumor and stimulated nonconstitutive steroid-producing MA-10 mouse Leydig tumor cells. *Endocrinology* **128**, 1918–1926.
- Stoker, T. E., Laws, S. C., Guidici, D. L., and Cooper, R. L. (2000). The effect of atrazine on puberty in male Wistar rats: An evaluation in the protocol for the assessment of pubertal development and thyroid function. *Toxicol. Sci.* **58**, 50–59.
- Trentacoste, S. V., Friedmann, A. S., Youker, R. T., Breckenridge, C. B., and Zirkin, B. R. (2001). Atrazine effects on testosterone levels and androgen-dependent reproductive organs in peripubertal male rats. *J. Androl.* **22**, 142–148.

- U.S. Environmental Protection Agency (USEPA). (2007). *ToxCast Phase 1 Assays*. Available at: <http://www.epa.gov/ncct/toxcast/>. Accessed April 16, 2012.
- U.S. Environmental Protection Agency (USEPA). (2009). *Endocrine Disruptor Screening Program Test Guidelines: Steroidogenesis (Human Cell Line-H295R)*. Prevention Pesticides and Toxic Substances. USEPA docket ID: EPA-HQ-OPPT-2009-0576, document ID: EPA-HQ-OPPT-2009-0576-0011.
- U.S. Environmental Protection Agency (USEPA). (2011). *Endocrine Disruptor Screening Program for the 21st Century: The Incorporation of In Silico Models and In Vitro High Throughput Assays in the Endocrine Disruptor Screening Program (EDSP) for Prioritization and Screening [EDSP21 Work Plan]*. Office of Chemical Safety and Pollution Prevention, Washington, DC.
- Vandesompele, J., De Preter, K., Pattyn, F., Poppe, B., Van Roy, N., De Paep, A., and Speleman, F. (2002). Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biol.* **3**, 1–12.
- Vasta, V., Shimizu-Albergine, M., and Beavo, J. A. (2006). Modulation of Leydig cell function by cyclic nucleotide phosphodiesterase 8A. *Proc. Natl. Acad. Sci. U.S.A.* **103**, 19925–19930.
- Victor-Costa, A. B., Bandeira, S. M., Oliveira, A. G., Mahecha, G. A., and Oliveira, C. A. (2010). Changes in testicular morphology and steroidogenesis in adult rats exposed to Atrazine. *Reprod. Toxicol.* **29**, 323–331.
- Vinggaard, A. M., Christiansen, S., Laier, P., Poulsen, M. E., Breinholt, V., Jarfelt, K., Jacobsen, H., Dalgaard, M., Nellemann, C., and Hass, U. (2005). Perinatal exposure to the fungicide prochloraz feminizes the male rat offspring. *Toxicol. Sci.* **85**, 886–897.
- Vinggaard, A. M., Hass, U., Dalgaard, M., Andersen, H. R., Bonefeld-Jorgensen, E., Christiansen, S., Laier, P., and Poulsen, M. E. (2006). Prochloraz: An imidazole fungicide with multiple mechanisms of action. *Int. J. Androl.* **29**, 186–192.
- Walsh, L. P., McCormick, C., Martin, C., and Stocco, D. M. (2000). Roundup inhibits steroidogenesis by disrupting steroidogenic acute regulatory (StAR) protein expression. *Environ. Health Perspect.* **108**, 769–776.
- Wang, X., and Stocco, D. M. (1999). Cyclic AMP and arachidonic acid: A tale of two pathways. *Mol. Cell. Endocrinol.* **158**, 7–12.
- Welsh, M., Saunders, P. T., Fiskens, M., Scott, H. M., Hutchison, G. R., Smith, L. B., and Sharpe, R. M. (2008). Identification in rats of a programming window for reproductive tract masculinization, disruption of which leads to hypospadias and cryptorchidism. *J. Clin. Investig.* **118**, 1479–1490.
- Whitehead, S. A., and Rice, S. (2006). Endocrine-disrupting chemicals as modulators of sex steroid synthesis. *Best Pract. Res. Clin. Endocrinol. Metab.* **20**, 45–61.
- Wilson, V. S., Blystone, C. R., Hotchkiss, A. K., Rider, C. V., and Gray, L. E., Jr (2008). Diverse mechanisms of anti-androgen action: Impact on male rat reproductive tract development. *Int. J. Androl.* **31**, 178–187.
- Wohlfahrt-Veje, C., Main, K. M., and Skakkebaek, N. E. (2009). Testicular dysgenesis syndrome: Foetal origin of adult reproductive problems. *Clin. Endocrinol. (Oxf.)* **71**, 459–465.
- Yao, H. H., and Barsoum, I. (2007). Fetal leydig cells. In *The Leydig Cell in Health and Disease* (A. H. Payne and M. P. Hardy, Eds.). Humana Press Inc., Totowa, NJ.
- Yen, S. S., Llerena, O., Little, B., and Pearson, O. H. (1968). Disappearance rates of endogenous luteinizing hormone and chorionic gonadotropin in man. *J. Clin. Endocrinol. Metab.* **28**, 1763–1767.
- Zhang, B., Cox, S. B., McMurry, S. T., Jackson, W. A., Cobb, G. P., and Anderson, T. A. (2008). Effect of two major N-nitroso hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) metabolites on earthworm reproductive success. *Environ. Pollut.* **153**, 658–667.
- Zhang, B., Freitag, C. M., Canas, J. E., Cheng, Q., and Anderson, T. A. (2006). Effects of hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) metabolites on cricket (*Acheta domesticus*) survival and reproductive success. *Environ. Pollut.* **144**, 540–544.
- Zorrilla, L. M., Gibson, E. K., Jeffay, S. C., Crofton, K. M., Setzer, W. R., Cooper, R. L., and Stoker, T. E. (2009). The effects of triclosan on puberty and thyroid hormones in male Wistar rats. *Toxicol. Sci.* **107**, 56–64.